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Title of Article

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Key Points:

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- RSTF

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- Source Inversion using RSTF and Teleseismic data

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- Compare slip models from RSTF and Teleseismic data

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Abstract

Will be done Later.

Plain Language Summary

Will be done Later.

1 Introduction**1.1 Elastic Rebound Theory and Earthquake Cycles on Transform Faults**

- Application of elastic rebound theory and seismic gap hypotheses to stress accumulation and release along fault boundaries.
- Shortcomings of the elastic rebound theory in explaining complex earthquake cycles, especially on oceanic transform faults.

1.2 Seismotectonic Setting of our study area and the fault structure

- Tectonic framework of the plate boundary, quasi-periodic seismicity, fault strike, and plate motion. (strike/dip/rake, slip rate, etc.)
- Structural characteristics of the fault plane: non-vertical, complex geometry.
- Overview of the 2015 and 2025 earthquake sequence occurring along the same fault system within a 10-year span. (figure on the seismicity and fault structure)

1.3 Unresolved Questions and Study Objectives

- In this slowly moving transform fault system, the 2015 and 2025 earthquakes occurred within a short time span of 10 years which is unexpected. This contradicts the natural hypothesis of stress loading and release along the fault system.
- Are the slip distributions of the 2015 and 2025 earthquakes similar or different? If different, what are the implications for stress accumulation and release along the fault system?

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2 Data and Methodology**2.1 Which methods are applied to investigate our problem**

- RSTF deconvolution.
- Teleseismic phase picking and preparation of the data for inversion.
- Finite-fault kinematic inversion using teleseismic and RSTF data to get slip distribution for both the 2015 and 2025 earthquakes.
- Details of these methods will be provided in the supporting information section.

2.2 How the method is applied***2.2.1 Data Selection and Teleseismic Preprocessing for Inversion***

- Continuous three-component waveform time series and corresponding metadata were retrieved for the 2015 and 2025 earthquakes from the GEOSCOPE database from the networks: IU, II, GT, G, GE, DK, CN.
- For RSTF deconvolution, we retrieved data for a radius of 40 degrees around the epicentres of the 2015 and 2025 earthquakes.

- 47 • For RSTF analysis, we selected the same earthquake which happened on 05/04/2025
- 48 as our EGF. Why this earthquake was selected as EGF will be explained in the
- 49 supporting information section.
- 50 • Instrument response was also removed to maintain the homogeneity of the data.
- 51 • For teleseismic phase picking, we selected data for a radius of 80 degrees to avoid
- 52 upper-mantle triplications and core reflections.
- 53 • The teleseismic P- and S-wave data were windowed for 70 and 115 s, respectively,
- 54 with arrival-time uncertainties of 2.5 and 7 s. The waveforms were processed us-
- 55 ing low- and broadband smoothing periods of 33 and 8 s, respectively, and a high-
- 56 frequency cutoff of 0.005 Hz was applied.

57 ***2.2.2 Finite-fault kinematic inversion***

58 ***Fault Plane Geometry***

59 **2015 earthquake**

- 60 • Strike: 98°
- 61 • Initial dip: 75°
- 62 • Initial rake: 175°
- 63 • Maximum rake deviation: $\pm 30^\circ$
- 64 • Maximum dip deviation: $\pm 30^\circ$
- 65 • Fault length: 198 km
- 66 • Fault width: 30 km
- 67 • Fault-centre coordinates: $(-77.24, 10.86, 18)$ km
- 68 • Hypocenter: subfault 96

69 **2025 earthquake**

- 70 • Strike: 98°
- 71 • Initial dip: 80°
- 72 • Initial rake: 175°
- 73 • Maximum rake deviation: $\pm 30^\circ$
- 74 • Maximum dip deviation: $\pm 30^\circ$
- 75 • Fault length: 200 km
- 76 • Fault width: 32 km
- 77 • Fault-centre coordinates: $(-63.47, 8.22, 19)$ km
- 78 • Hypocenter: subfault 71

79 ***Fault Discretization***

80 **2015 earthquake**

- 81 • Subfault dimensions: 6×6 km
- 82 • Fault dimensions: 198×30 km
- 83 • Hypocenter subfault: 96

84 **2025 earthquake**

- 85 • Subfault dimensions: 8×8 km
- 86 • Fault dimensions: 200×32 km
- 87 • Hypocenter subfault: 71

Inverted Source Parameters

The finite-fault inversion solves for the spatial and temporal characteristics of the rupture. The principal source parameters include: slip amplitude, rake, spatial variation in fault dip

Inversion Configuration

The principal numerical parameters used in the finite-fault inversion include:

- Rigidity: 4×10^{11} Pa
- Initial rupture velocity: 3.5 km/s

The complete inversion parameter configuration is provided in the Supporting Information.

Final Inversion Products

For each earthquake, the inversion provides:

- Slip distribution
- Rake distribution
- Rupture-time distribution
- Dip distribution
- Synthetic waveforms
- Observed–synthetic waveform misfit

2.2.3 Comparison of the 2015 and 2025 earthquake slip distributions

- Slip distributions are plotted for both the 2015 and 2025 earthquakes. The slip distributions are compared to understand the similarities and differences in the rupture processes of the two events.
- Contours of the slips from both earthquakes are plotted in a single figure to visualize the spatial relationship between the two slip distributions.
- To plot the contours we distributed the absolute slip values into 5 equal intervals starting from minimum slip of 8cm and plotted the contours for both earthquakes in a single figure.

2.2.4 Comparison between source time functions of the 2015 and 2025 earthquakes

- The STFs are plotted for both earthquakes and the differences in the duration, peak amplitude, and overall shape of the STFs are analyzed.

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: <https://www.agu.org/Publish> with AGU/Publish/Author Resources/Data for Authors

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